Haniyeh Zahra Budaqi

Email | LinkedIn | GitHub | Google Scholar

Embedded software engineer with real-time firmware experience (C/C++, FreeRTOS) across embedded and robotic systems, and a research background in spacecraft GNC and astrodynamics for autonomous rendezvous and proximity operations.

Education

B.Sc. Mechanical Engineering Nov 2020 – Jul 2025
Sharif University of Technology *Tehran, Iran*
- **GPA:** 3.44/4.00 (16.38/20.00)
- **Focus Areas:** Dynamics, Robotics, Control Systems; coursework in Modern Control Theory and Orbital Mechanics

Technical Skills

- **Embedded & Real-Time Systems:** STM32, ESP32, AVR; FreeRTOS; SPI, I2C, UART, DMA, CAN, BLE; PCB design (Altium)
- **Programming & Version Control:** C/C++, Python, MATLAB/Simulink, Git
- **Control & Simulation:** Spacecraft attitude dynamics, nonlinear control, PID control, system modeling, STK
- **Languages:** English (Fluent – IELTS Academic 8.5/9.0, CEFR C2), French (A2, actively studying)

Engineering Experience

Embedded Software Engineer
Lian Battery, Tehran, Iran Jan 2026 – Present
- Develop real-time C/C++ firmware (FreeRTOS, ESP32-S3) with CAN-based subsystem communication interfaces.

MG Adak, Tehran, Iran Apr 2025 – Jan 2026
- Developed modular STM32 firmware for commercial IoT systems, focused on reliability and scalability.

Electronics Assistant Oct 2025 – Jan 2026
Sharif Social Robotics Lab *Tehran, Iran*
- Designed 2-layer PCBs and implemented BLE-based communication for real-time telemetry and control on robotic platforms.

Electronics Engineering Intern Nov 2024 – Jul 2025
Nira System Pouya *Tehran, Iran*
- Completed specialized training in Embedded C and PCB design through hands-on integration projects.

Selected Projects

Attitude Dynamics and Control of a Nano-Satellite Orbiting Mars | **MATLAB, STK** Mar 2026 – Jun 2026
- Developed a nonlinear attitude simulator and Lyapunov-based controller (MRPs) for a Mars-orbiting nano-satellite, validated via STK.

Publications

"Projection-Constrained Generative Initialization of NRHO-DRO Transfers in the Earth-Moon CR3BP" Sep 2026
***H. Z. Budaqi**, P. Latifiyan, M. Arsalan* *Submitted to Acta Astronautica*
Access the manuscript

"Postural Responses to Mediolateral Rotations" Nov 2025
***H. Z. Budaqi**, A. Mojibi, S. Behzadipour* *ICBME Conference*
10.1109/ICBME68496.2025.11392127

Courses & Certifications

- Spacecraft Dynamics and Control Specialization (CU Boulder) • Intro to MATLAB Programming (Vanderbilt) • CS50 (Harvard)